Networks and Complexity

Solution 17-4

This is an example solution from the forthcoming book Networks and Complexity. Find more exercises at https://github.com/NC-Book/NCB

Ex 17.4: Linear manifold example [Lvl. 2]

Use eigendecomposition to find the invariant manifolds of the steady state in the following linear system:

$$\begin{aligned} \dot{x} &= 2x + 4y \\ \dot{y} &= x - y, \end{aligned}$$

<u>*Solution*</u>

Because the system is linear we know that the trajectories have the form

$$\boldsymbol{x}(t) = c_1 \boldsymbol{v_1} \mathrm{e}^{\lambda_1 t} + c_2 \boldsymbol{v_2} \mathrm{e}^{\lambda_2 t} \tag{1}$$

hence we need to compute the eigenvectors and eigenvalues of the Jacobian matrix

$$\mathbf{J} = \begin{pmatrix} 2 & 4 \\ 1 & -1 \end{pmatrix}. \tag{2}$$

We find the characteristic polynomial as

$$0 = \left| \begin{pmatrix} 2-\lambda & 4 \\ 1 & -1-\lambda \end{pmatrix} \right| \tag{3}$$

which yields

$$\begin{aligned} 0 &= (2-\lambda)(-1-\lambda) - 4 && (4) \\ 0 &= \lambda^2 - \lambda - 6 && (5) \\ 0 &= (\lambda - 1/2)^2 - 1/4 - 6 && (6) \\ 0 &= (\lambda - 1/2)^2 - 25/4 && (7) \\ 0 &= \lambda - 1/2 \pm 5/2 && (8) \\ & && (9) \end{aligned}$$

Hence we find the eigenvalues

$$\lambda_1 = 3 \qquad \lambda_2 = -2 \tag{10}$$

We now determine the eigenvectors in the usual way. First for the eigenvalue $\lambda_1 = 3$:

$$\begin{pmatrix} 2 & 4 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ a \end{pmatrix} = \begin{pmatrix} 3 \\ 3a \end{pmatrix} \tag{11}$$

The lower row reads

$$\begin{aligned} 1 - 1a &= 3a && (12) \\ a &= 1/4 && (13) \end{aligned}$$

multiplying the whole vector by 4 we write

$$\boldsymbol{v_1} = \begin{pmatrix} 4 \\ 1 \end{pmatrix} \tag{14}$$

Using the same approach for the second eigenvector yields

$$\begin{pmatrix} 2 & 4 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ a \end{pmatrix} = \begin{pmatrix} -2 \\ -2a \end{pmatrix} \tag{15}$$

and hence

$$1 - a = -2a \tag{16}$$
$$a = -1 \tag{17}$$

which gives us the eigenvector

$$\boldsymbol{v_2} = \begin{pmatrix} 1 \\ -1 \end{pmatrix} \tag{18}$$

We now know that trajectories have the form

$$\boldsymbol{x}(t) = c_1 \begin{pmatrix} 4 \\ 1 \end{pmatrix} \mathrm{e}^{3t} + c_2 \begin{pmatrix} 1 \\ -1 \end{pmatrix} \mathrm{e}^{-2t} \tag{19}$$

If our initial value is such that $c_1 = 0$ we are on a trajectory

$$\boldsymbol{x}(t) = c_2 \begin{pmatrix} 1 \\ -1 \end{pmatrix} \mathrm{e}^{-2t} \tag{20}$$

that approaches the state (0,0) and hence we are on the stable manifold. We can also write this as

$$x = c_2 \mathrm{e}^{-2t} \tag{21}$$
$$y = -c_2 \mathrm{e}^{-2t} \tag{22}$$

for all values of c_2 and t. Substituting the first equation into the second we find

$$y = -x \tag{23}$$

which describes the manifold that we are looking for. In the same way the unstable manifold consists of the points visited by trajectories for which $c_2 = 0$, so

$$\boldsymbol{x}(t) = c_1 \begin{pmatrix} 4 \\ 1 \end{pmatrix} \mathrm{e}^{3t} \tag{24}$$

which means

$$x = 4c_1 \mathrm{e}^{3t} \tag{25}$$
$$y = c_1 \mathrm{e}^{3t} \tag{26}$$

such that

$$y = \frac{1}{4} x \tag{27}$$

is the unstable manifold we are looking for.